

Git

Git is a distributed version control system (DVCS) used to track changes in files and source code. It helps developers manage projects, collaborate with teams, and maintain different versions of their work.

Features of Git

- Tracks file changes.
- Supports team collaboration.
- Maintains version history.
- Allows branching and merging.
- Fast and efficient.

Basic Git Commands

Command	Purpose
<code>git init</code>	Initialize a new Git repository.
<code>git clone <url></code>	Copy a remote repository.
<code>git status</code>	Show the current status of files.
<code>git add <file></code>	Add a file to the staging area.
<code>git add .</code>	Add all files to the staging area.
<code>git commit -m "message"</code>	Save staged changes with a message.
<code>git log</code>	Display commit history.
<code>git branch</code>	List all branches.
<code>git switch <branch></code>	Switch to another branch.
<code>git merge <branch></code>	Merge a branch into the current branch.
<code>git push origin main</code>	Upload local commits to the remote repository.
<code>git pull</code>	Download and merge changes from the remote repository.
<code>git fetch</code>	Download changes without merging.
<code>git diff</code>	Show differences between file versions.
<code>git restore <file></code>	Discard changes in a file.
<code>git stash</code>	Temporarily save uncommitted changes.

Conclusion: Git is an essential tool for version control that helps developers track changes, collaborate efficiently, and manage software projects effectively.

SCREENSHOTS:

```

cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git status
On branch main
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    ass2/index.html
        deleted:    ass2/script.js
        deleted:    ass2/style.css

no changes added to commit (use "git add" and/or "git commit -a")
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git add .
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git commit -m "new git command practice"
[main e1696fa] new git command practice
 3 files changed, 472 deletions(-)
 delete mode 100644 ass2/index.html
 delete mode 100644 ass2/script.js
 delete mode 100644 ass2/style.css
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git status
On branch main
nothing to commit, working tree clean
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 616 bytes | 616.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/AayushDSE/College.git
   257c58f..e1696fa  main -> main
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git log --oneline
e1696fa (HEAD -> main, origin/main) new git command practice
5268d03 new git command
fe5a81f changed folder name
257c58f added basic js functionalities
d2b2516 add login page to the website
4eae253 added design to dashboard and login
5d7b108 ass 2 and 3
92f8175 add design to the website and basic functionality
403c307 created basic structure of the website
9f7cfea Expense Tracker project started
a2bae6d Changed folder names
6ff1815 Merge pull request #1 from AayushDSE/extra
8972031 (extra) Practiced basic js code
769216a student database project created
56e9769 netflix css done
303b943 created javascript file
0de0a6e Changes title of index file
9f79b41 created completely new code
da2cb7c changes in code
830ef32 created netflix frontend clone
fe2d485 Merge branch 'main' of https://github.com/AayushDSE/College into main
79bdc0e movies code

```

```

a30fdbd New folder created
4cb7047 Initial commit
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git remote -v
origin  https://github.com/AayushDSE/College.git (fetch)
origin  https://github.com/AayushDSE/College.git (push)
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git config --list
user.name=Aayush
user.email=aayush.warkhede@gmail.com

core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
remote.origin.url=https://github.com/AayushDSE/College.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
cclab-01@parth-cha-pc:~/Aayush(VSEC)$ git mv ass1 ass1-basics

```